

1. Administrative & Study Identification

Full Study Title: A Phase 2b, Randomized, Double-Blind, Placebo-Controlled Study to Evaluate the Efficacy and Safety of Tulisokibart in Participants With Radiographic Axial Spondyloarthritis (Ankylosing Spondylitis) **Short Title/Acronym:** MK-7240-013 **Protocol Number:** MK-7240-013 **NCT Number:** NCT07133633 **Sponsor Name:** Merck Sharp & Dohme LLC **Investigational Product:** Tulisokibart (MK-7240) **Phase of Development:** Phase 2b **Medical Monitor:** Clinical Operations Lead, Merck Sharp & Dohme LLC

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of tulisokibart (low, medium, and high doses) compared to placebo in achieving an Assessment of SpondyloArthritis international Society 40 (ASAS 40) response at Week 16. **Secondary Objectives:** To evaluate the long-term safety, tolerability, and sustained symptomatic improvement of spinal inflammation, mobility, and joint pain over a 16-week controlled period followed by a 124-week long-term extension.

2.2 Background & Rationale To address the unmet medical need in patients with radiographic axial spondyloarthritis (r-axSpA, ankylosing spondylitis) who experience active disease despite standard treatments (e.g., NSAIDs). Tulisokibart is a novel anti-TL1A monoclonal antibody that inhibits inflammatory pathways and reduces tissue fibrosis, presenting a unique therapeutic mechanism for targeting immune-mediated inflammatory diseases.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized (Parallel Assignment)

3.2 Planned Enrollment & Duration Target \$N\$: Targeting a subset of approx. 200 participants (part of a broader 640-patient global clinical program across three indications) **Number of Sites:** ~88 locations globally **Estimated Study Start:** August 2025 **Estimated Primary Completion:** Q4 2027 (Results projected ~2028)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Has a clinical diagnosis of axial spondyloarthritis (axSpA) and meets the ASAS classification criteria for axSpA, including ≥ 3 months of back pain with age at symptom onset < 45 years. 2. Meets the radiographic criterion of the modified New York criteria for ankylosing spondylitis (AS) as determined by central reading at Screening. 3. Has active disease at Screening and Randomization, along with a history of inadequate response (IR) or intolerance to nonsteroidal anti-inflammatory drugs (NSAIDs).

4.2 Exclusion Criteria 1. Has any arthritis with onset before age 17 years or current diagnosis of inflammatory joint disease other than r-axSpA (such as rheumatoid arthritis, systemic lupus erythematosus, psoriatic arthritis, systemic sclerosis, etc.). 2. Has a history of cancer (except fully treated non-melanoma skin cancers or cervical carcinoma in situ) within the last 5 years. 3. Has any active infection, or known allergies/hypersensitivity to tulusokibart or its excipients.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Participants are randomized to receive low-dose, medium-dose, or high-dose tulusokibart, or a matched placebo. At Week 16, participants on placebo and low-dose tulusokibart will be rerandomized to receive either the medium or high dose. **Route of Administration:** Subcutaneous injection. **Duration of Treatment:** 16-week Placebo-controlled Period followed by a 124-week Long-term Extension (LTE), totaling up to 140 weeks of exposure.

5.2 Concomitant & Prohibited Therapy Permitted: Standard background therapies (e.g., stable doses of NSAIDs) as dictated by the protocol. **Prohibited:** Concurrent use of other investigational therapies or non-approved biologic DMARDs for ankylosing spondylitis.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Radiographic Central Reading
Baseline	Day 0	Randomization, First Dose, Safety Labs
Interim	Week 16 $\pm$ 7	Efficacy Assessment (ASAS 40), Rerandomization into LTE Phase
End of Study	Week 140	Final Vital Signs, Long-term Safety Audit, End of LTE Phase

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Proportion of participants achieving an ASAS 40 response at Week 16. **Measurement Method:** Clinical assessment evaluated by the Assessment of SpondyloArthritis international Society (ASAS) criteria.

7.2 Secondary & Exploratory Endpoints * Long-term safety and tolerability (Adverse Event tracking over 140 weeks). * Changes from baseline in spinal inflammation, mobility, and symptom severity (e.g., using established imaging parameters and scoring systems).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring through patient reporting and routine clinical evaluation up to Week 140. **Serious Adverse Events (SAEs):** Standard reporting timelines within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Independent safety review committee designated by Merck Sharp & Dohme LLC.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy measures (ASAS 40). Safety Set for all participants receiving at least one dose of the investigational product. **Sample Size Justification:** Adequately powered to detect a statistically significant distinction in ASAS 40 response rates between active tulisokibart dosage arms and the placebo arm at Week 16. **Handling of Missing Data:** Non-Responder Imputation (NRI) for categorical efficacy endpoints; mixed-effect models for repeated measures (MMRM) for continuous endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine on-site monitoring and centralized data review throughout the global sites. **Audit Trail:** eCRF locking mechanisms and rigorous data clarification tracking per Merck's clinical standards.

11. Study Identifiers & Governance

Primary ID: MK-7240-013 **Registry Number:** NCT07133633 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** A Clinical Study of Tulisokibart (MK-7240) to Treat Radiographic Axial

Spondyloarthritis (MK-7240-013) **Official Regulatory Title:** A Phase 2b, Randomized, Double-Blind, Placebo-Controlled Study to Evaluate the Efficacy and Safety of Tulasokibart in Participants With Radiographic Axial Spondyloarthritis (Ankylosing Spondylitis)

12. Clinical Framework (r-axSpA Specific)

Primary Condition: Radiographic Axial Spondyloarthritis (Ankylosing Spondylitis) **Diagnosis Threshold:** Meets ASAS criteria & modified New York criteria for AS via central reading **Medical Methodology:** Anti-TL1A Monoclonal Antibody Therapy **Optimization Target:** Achievement of ASAS 40 response

13. Study Procedures & Methodology

Management Pathway: Biological therapy protocol (Tulasokibart vs. Placebo) **Education Protocol (HCP):** Standardized training for radiographic assessments and ASAS scoring **Education Protocol (Patient):** Protocol orientation, symptomatic logging, and treatment adherence guidance **Quality Audit Mechanism:** Centralized radiographic reading to confirm eligibility and regular clinical monitoring

14. Observation & Follow-Up Schedule

Total Duration: 140 weeks (16 weeks placebo-controlled + 124-week long-term extension) **Onsite Visit Cadence:** Baseline, Week 16, and periodic LTE phase visits **Remote Monitoring Frequency:** Regular safety touchpoints as defined by the protocol **Checklist Review Interval:** Routine adverse event and concomitant medication review corresponding to dosing visits

15. Sign-off & Approval

	Role		Name		Signature		Date			:---		:---		:---		:---	
	Principal Investigator				[Insert Name]			____		[DD-MMM-YYYY]							
	Clinical Operations Lead				[Insert Name]			____		[DD-MMM-YYYY]							
	Lead Statistician				[Insert Name]			____		[DD-MMM-YYYY]							